

MOMENTO

Location-Based Time Capsule Application

TEST REPORT

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1. Introduction

1.1 Definitions of the abbreviations

Abbreviation	Explanation
Momento	Location-based time capsule mobile application
API	Application Programming Interface
GPS	Global Positioning System
JWT	JSON Web Token
FCM	Firebase Cloud Messaging
S3	Amazon Simple Storage Service
PostGIS	Spatial database extension for PostgreSQL
UUID	Universally Unique Identifier
UI	User Interface
iOS	Apple mobile operating system
Android	Google mobile operating system

1.2 Document purpose

This document describes the test report for the Momento location-based time capsule application (hereinafter "Momento").

The document is intended for the parties involved in testing and maintenance of the software:

- Instructors and evaluators responsible for accepting and evaluating the project deliverables.
- Technical members of the development team responsible for implementation, design, and maintenance.

Momento allows users to "drop" digital time capsules — containing text messages and photos — at a specific GPS location. Other users who physically travel to that location can then discover and unlock the capsule, earning points in the process.

1.3 Related Documents

This test report has been developed based on the following documents:

- [1] Software Architecture Description (ARCHITECTURE.md)
- [2] Environment Setup Guide (ENVIRONMENT_SETUP.md)
- [3] Project README (README.md)
- [4] Database Schema (database/schema.sql)

1.4 General requirements for test execution

In order to test the operation of the system, the tester needs:

- A physical or virtual mobile device (iOS or Android) with the Expo Go application installed, or a built Expo development client.
- A running backend instance (Spring Boot) accessible over the network.
- A configured PostgreSQL database with the PostGIS extension and the applied schema.
- A valid Firebase project with Email/Password authentication enabled.
- A configured AWS S3 bucket for media storage.
- At least two registered user accounts for testing self-discovery blocking scenarios.
- A device with GPS / location services enabled.

2. Functions that need testing

No.	Description	Requirement
1	User Authentication	MOMENTO-1
2	Capsule Creation	MOMENTO-2
3	Map Discovery (Nearby Capsules)	MOMENTO-3
4	Capsule Unlock (Proximity Gate)	MOMENTO-4
5	Discovery History	MOMENTO-5
6	Dropped Capsules Management	MOMENTO-6
7	User Profile	MOMENTO-7

3. Test Scenarios

3.1 User Authentication

3.1.1 Checking that required fields are validated on login

- Open the Login screen.
- Leave the Email and Password fields empty.
- Press the "Sign in" button.

Expected result: An alert is displayed with the message "Missing fields – Please enter your email and password."

Test result: Passed

3.1.2 Checking that an incorrect email or password is rejected

- Open the Login screen.
- Enter the value "wrong@test.com" in the Email field.
- Enter the value "wrongpassword" in the Password field.

- Press the "Sign in" button.

Expected result: An alert is displayed with the message "Login failed" and the Firebase error reason.

Test result: Passed

3.1.3 Successful user login

- Open the Login screen.
- Enter a valid registered email address in the Email field.
- Enter the correct password in the Password field.
- Press the "Sign in" button.

Expected result: The user is authenticated and automatically navigated to the main application screen (Map tab). The session persists after closing and reopening the app.

Test result: Passed

3.1.4 Checking that required fields are validated on registration

- Open the Register screen via the "Create account" button.
- Leave the Email and Password fields empty.
- Press the "Register" button.

Expected result: An alert is displayed with the message "Missing fields – Please fill in all fields."

Test result: Passed

3.1.5 Checking password length validation on registration

- Open the Register screen.
- Enter a valid email address.
- Enter a password shorter than 6 characters (e.g. "abc").
- Press the "Register" button.

Expected result: An alert is displayed with the message "Weak password – Password must be at least 6 characters."

Test result: Passed

3.1.6 Successful user registration

- Open the Register screen.
- Enter a new, unused email address.
- Enter a valid password of at least 6 characters.
- Press the "Register" button.

Expected result: The account is created in Firebase and the user is automatically navigated to the main application screen.

Test result: Passed

3.1.7 User sign-out with confirmation

- Navigate to the Profile tab.
- Press the "Sign out" button.
- A confirmation dialog appears — press "Sign out" to confirm.

Expected result: The user is signed out and the Login screen is displayed.

Test result: Passed

3.1.8 Cancelling sign-out

- Navigate to the Profile tab.
- Press the "Sign out" button.
- A confirmation dialog appears — press "Cancel".

Expected result: The dialog dismisses and the user remains signed in on the Profile screen.

Test result: Passed

3.2 Capsule Creation

Sign in with a valid user account. Navigate to the Create tab.

3.2.1 Blocking capsule creation without GPS fix

- Open the Create screen before location has been acquired.
- Press the "Drop capsule" button immediately.

Expected result: An alert is displayed with the message "Location required – Waiting for GPS fix. Try again in a moment."

Test result: Passed

3.2.2 Blocking creation of an empty capsule

- Open the Create screen after a GPS fix has been obtained.
- Leave the Message field empty and do not select a photo.
- Press the "Drop capsule" button.

Expected result: An alert is displayed with the message "Empty capsule – Add a message or photo before dropping."

Test result: Passed

3.2.3 Creating a text-only capsule

- Open the Create screen.
- Wait for the GPS coordinates to appear at the top of the screen.
- Enter a message in the Message field (e.g. "Hello from this spot!").
- Press the "Drop capsule" button.

Expected result: An alert is displayed confirming "Capsule dropped! Your capsule was saved at your current location." The text fields are cleared after successful creation.

Test result: Passed

3.2.4 Selecting a photo from the library

- Open the Create screen.
- Press the "Choose photo" button.
- Grant media library permissions when prompted.
- Select a photo from the library.

Expected result: The button label changes to "Photo selected — tap to change" indicating a photo has been attached.

Test result: Passed

3.2.5 Creating a capsule with a photo

- Open the Create screen and wait for GPS.
- Select a photo using the "Choose photo" button.
- Optionally add a text message.
- Press the "Drop capsule" button.

Expected result: The capsule is created with the attached photo uploaded to S3. A success alert is displayed and both fields are cleared.

Test result: Passed

3.2.6 Blocking duplicate capsule at the same location on the same day

- Drop a capsule at the current location.
- Without moving, attempt to drop another capsule at the exact same coordinates.

Expected result: The server returns an error and an alert is displayed indicating a duplicate drop constraint was violated.

Test result: Passed

3.3 Map Discovery

Sign in with a valid user account. Navigate to the Map tab.

3.3.1 Displaying nearby capsules

- Navigate to the Map tab.
- Wait for the GPS fix and the capsule list to load.

Expected result: A list of nearby capsules is displayed. Each entry shows the distance in metres and the GPS coordinates of the capsule.

Test result: Passed

3.3.2 Displaying empty state when no capsules are nearby

- Navigate to the Map tab from a location with no dropped capsules within 50 metres.

Expected result: An empty state is shown with the message "No capsules nearby yet. Drop one and come back!"

Test result: Passed

3.3.3 Displaying GPS error when location is denied

- Deny location permissions on the device.
- Open the Map tab.

Expected result: An error message is shown explaining that location access was denied.

Test result: Passed

3.3.4 Refreshing the capsule list

- Navigate to the Map tab with at least one nearby capsule.
- Pull down on the capsule list to trigger a refresh, or press the "Refresh" button.

Expected result: The list reloads and displays the current nearby capsules. The refresh indicator is visible during loading and disappears when complete.

Test result: Passed

3.3.5 Navigating to capsule detail

- Navigate to the Map tab.
- Tap on a capsule entry in the list.

Expected result: The Capsule Detail screen opens showing the distance of the selected capsule and an "Try unlock" button.

Test result: Passed

3.4 Capsule Unlock

Sign in with a valid user account. Navigate to the Map tab and open a nearby capsule.

3.4.1 Successful proximity unlock

- Navigate to the Capsule Detail screen for a capsule within the unlock radius (50 metres).
- Press the "Try unlock" button.

Expected result: The capsule is unlocked. The screen shows the text content and any attached media. Points are awarded and the discovery is recorded.

Test result: Passed

3.4.2 Unlock blocked outside proximity radius

- Navigate to the Capsule Detail screen for a capsule that is more than 50 metres away.
- Press the "Try unlock" button.

Expected result: An alert is displayed with the message "Unlock failed" indicating the user is not close enough to unlock.

Test result: Passed

3.4.3 Unlock blocked without GPS fix

- Open the Capsule Detail screen when location is unavailable.
- Press the "Try unlock" button.

Expected result: An alert is displayed with the message "Location unavailable – Cannot unlock without a GPS fix."

Test result: Passed

3.4.4 Self-discovery blocked

- Drop a capsule as User A at a specific location.
- While still signed in as User A, navigate to the Map tab and open that same capsule.
- Press the "Try unlock" button.

Expected result: An alert is displayed indicating the server blocked the unlock because users cannot discover their own capsules.

Test result: Passed

3.4.5 Duplicate unlock blocked

- Unlock a capsule as User B.
- Without navigating away, attempt to press "Try unlock" again, or return to the same capsule and try again.

Expected result: The server rejects the second unlock attempt. An alert is displayed indicating the capsule was already discovered by this user.

Test result: Passed

3.4.6 Viewing media after unlock

- Successfully unlock a capsule that contains a photo.
- The unlocked view shows "View media >" link.
- Tap the link.

Expected result: The device opens the pre-signed S3 URL in the browser or media viewer, displaying the attached photo.

Test result: Passed

3.5 Discovery History

Sign in with a valid user account. Navigate to the History tab.

3.5.1 Displaying discovery history

- Navigate to the History tab after having previously unlocked at least one capsule.

Expected result: A list of past discoveries is displayed. Each entry shows the discovery date and the points awarded for that discovery.

Test result: Passed

3.5.2 Displaying empty state when no discoveries exist

- Sign in with a new user account that has not yet unlocked any capsules.
- Navigate to the History tab.

Expected result: An empty state is shown with the message "No discoveries yet. Find capsules near you on the Map tab!"

Test result: Passed

3.5.3 Refreshing the history list

- Navigate to the History tab.
- Pull down on the list to trigger a refresh.

Expected result: The list reloads and displays the latest discovery history. The refresh indicator is visible during loading.

Test result: Passed

3.6 Dropped Capsules Management

Sign in with a valid user account. Navigate to the My Capsules tab.

3.6.1 Displaying dropped capsules list

- Navigate to the My Capsules tab after having previously dropped at least one capsule.

Expected result: A list of the user's own capsules is displayed. Each entry shows the capsule status (ACTIVE / EXPIRED) and its expiry date.

Test result: Passed

3.6.2 Displaying empty state when no capsules have been dropped

- Sign in with a new user account that has not yet dropped any capsules.
- Navigate to the My Capsules tab.

Expected result: An empty state is shown with the message "No capsules dropped yet."

Test result: Passed

3.6.3 Delete confirmation dialog

- Navigate to the My Capsules tab with at least one capsule.
- Press the "Delete" button next to a capsule.

Expected result: A confirmation dialog is shown with the message "Delete capsule – This cannot be undone. Are you sure?" with Cancel and Delete options.

Test result: Passed

3.6.4 Cancelling a deletion

- Press the "Delete" button next to a capsule.
- In the confirmation dialog, press "Cancel".

Expected result: The dialog dismisses and the capsule remains in the list unchanged.

Test result: Passed

3.6.5 Successfully deleting a capsule

- Press the "Delete" button next to a capsule.
- In the confirmation dialog, press "Delete" to confirm.

Expected result: The capsule is permanently removed from the database (and from S3 if it contained media). The item disappears from the list without requiring a manual refresh.

Test result: Passed

3.6.6 Refreshing the capsules list

- Navigate to the My Capsules tab.
- Pull down on the list to trigger a refresh.

Expected result: The list reloads with the current state of the user's dropped capsules.

Test result: Passed

3.7 User Profile

Sign in with a valid user account. Navigate to the Profile tab.

3.7.1 Displaying profile statistics

- Navigate to the Profile tab.

Expected result: The profile card displays the user's username (or email if no username is set), total points, number of capsules dropped, and number of capsules discovered.

Test result: Passed

3.7.2 Profile statistics accuracy after dropping a capsule

- Note the current "Dropped" count on the Profile tab.
- Navigate to the Create tab and drop a new capsule.
- Return to the Profile tab and pull to refresh.

Expected result: The "Dropped" count is incremented by 1 compared to the value noted before.

Test result: Passed

3.7.3 Profile statistics accuracy after discovering a capsule

- Note the current "Discovered" count and "Points" total on the Profile tab.
- Unlock a nearby capsule as a different user (User B discovering User A's capsule).
- Navigate to the Profile tab and pull to refresh.

Expected result: The "Discovered" count is incremented by 1 and the "Points" total increases by the awarded points.

Test result: Passed

3.7.4 Refreshing the profile

- Navigate to the Profile tab.
- Pull down on the screen to trigger a refresh.

Expected result: The profile statistics reload and display up-to-date values.

Test result: Passed